

Paper Replications: Phase 1: Foundational Machine Learning Models

This project is a learn-by-doing exercise designed to build a practical understanding of foundational machine learning models through replication-style problems. The objective is not only to run Python code, but to understand how each model works, why it is used, how its outputs are interpreted, and how it can be applied to real problems in Financial Engineering, Economics, and Climate Economics/Risk.

The broader project is titled **Paper Replications** because each exercise is inspired by, or directly connected to, an existing research paper, research tradition, or publicly available empirical dataset. Instead of learning models only through abstract theory, this project uses real datasets and research-style questions to understand how machine learning and econometric models are applied in practice.

Phase 1 focuses on five core models:

1. Linear Regression
2. Logistic Regression
3. Ridge Regression
4. LASSO Regression
5. Elastic Net Regression

For each model, the plan is to complete three problems with increasing difficulty: an easy problem, a medium problem, and a hard problem. The problems are selected from areas relevant to my academic and career interests, especially asset pricing, credit risk, macroeconomic prediction, climate risk, and climate economics.

Linear Regression

The first part of Phase 1 begins with Linear Regression. The three Linear Regression exercises move from a simple CAPM beta estimation, to a Fama-French multi-factor regression, and then to a panel-style climate economics problem examining the relationship between temperature shocks and GDP growth. This progression helps build an understanding of simple regression, multiple regression, fixed effects, and interaction terms.

The purpose of this phase is to develop practical modelling confidence. Each exercise includes data preparation, model estimation, interpretation of coefficients, evaluation of results, and a short research-style write-up. By the end of Phase 1, the aim is to have a portfolio of beginner-friendly but academically relevant replication exercises that demonstrate both coding ability and conceptual understanding of core machine learning models.

Table 1. *Overview of Problems, Research Anchors, and Datasets*

Level	Problem	Research Anchor	Dataset
Easy	Estimate CAPM beta and alpha for industry portfolios	CAPM / Fama-French asset-pricing tradition; Fama and French identify common market, size, and value factors in stock and bond returns. (Bauer College of Business)	Kenneth French: Fama/French 3 Factors + 10 or 49 Industry Portfolios
Medium	Compare CAPM vs Fama-French 3-factor model	Fama-French 1993	Kenneth French: 3 Factors + 25 Size/BM Portfolios or 49 Industry Portfolios
Hard	Estimate the impact of temperature shocks on GDP growth	Dell, Jones & Olken, <i>Temperature Shocks and Economic Growth</i> ; the public replication package includes climate and economic panel files. (American Economic Association)	Dell–Jones–Olken replication data; optionally World Bank CCKP + WDI

Note: All datasets are publicly available through the Kenneth R. French Data Library and the AEA/openICPSR replication archive.

1. CAPM Beta Estimation for an Industry Portfolio

Problem 1: Linear Regression Replication — Estimating the CAPM Beta of the U.S. Energy Industry Portfolio Stocks and Bonds

Research Anchor: Fama and French (1993), “Common Risk Factors in the Returns on Stocks & Bonds”

This exercise uses a CAPM-style linear regression to estimate the market beta of the U.S. Energy industry portfolio. The empirical setup is inspired by Fama and French’s factor-based asset-pricing framework, using monthly data from the Kenneth R. French Data Library. The Fama-French data library is widely used in empirical asset-pricing research, and Fama and French’s 1993 paper studies common risk factors in stock and bond returns.

What is the **market beta** of an industry portfolio?

A simple linear regression has this form:

$$y = \alpha + \beta x + \epsilon$$

In our CAPM problem:

$$R_i - R_f = \alpha + \beta(R_m - R_f) + \epsilon$$

Our main goal is to **estimate β** . Beta tells us how much an asset or portfolio moves when the market moves.

We need two files from Kenneth French’s Data Library:

Dataset A: Fama/French 3 Factors

Dataset B: 10 Industry Portfolios: This gives monthly returns for industries such as:

NoDur, Durbl, Manuf, Enrgy, HiTec, Telcm, Shops, Hlth, Utils, Other

For our first run, we will use *Enrgy*

The exact model we will estimate

$$Enrgy_t - RF_t = \alpha + \beta(MktRF_t) + \epsilon_t$$

In Python terms:

$y = Enrgy$ excess return

$x = Mkt$ excess return

In the regression output, focus on these parts:

- const: α – It tells us whether Energy earned extra return beyond what CAPM explains.
- Mkt-RF: β

- **R-squared** – This tells us how much of the variation in Energy excess returns is explained by the market.
- $P > |t|$ - It is the **p-Value**

Residual Plot helps answer: Where is the model making errors?

Residual = Actual value - Predicted value

- If residual is above zero: Energy performed better than the CAPM model predicted
- If residual is below zero: Energy performed worse than the CAPM model predicted

Large spikes tell us that something happened in Energy returns that the market factor alone could not explain.

Findings

- Market explains 53.9% of Energy excess return movement.
- Beta is statistically significant.
- Alpha is not statistically significant.
- Energy is moving **in the same direction as the market**, but slightly less strongly than the market. Over this full sample, the Energy industry portfolio appears slightly less market-sensitive than the overall market portfolio.
- The alpha is 0.1553% per month. But if we look at the p-value of alpha, which is = 0.222, is larger than 0.05. The alpha is positive, but **it is not statistically significant**. Therefore, we **do not have strong evidence** that the Energy portfolio earned abnormal excess returns beyond what is explained by market risk.
- Your R-squared is 0.539. About 53.9% of the variation in Energy industry excess returns is explained by movements in the market excess return. That is quite reasonable for a single-factor CAPM model. That remaining part may come from oil prices, geopolitical shocks, sector-specific news, supply shocks, interest rates, inflation, or other industry-specific factors.
- The p-value for Mkt-RF is shown as 0.000. This does not mean the p-value is exactly zero. It means it is extremely small, rounded to 0.000 by Python. So write it as $p < 0.001$. This means that the market excess return is a statistically significant predictor of Energy industry excess returns. Simply, the market matters a lot for explaining Energy returns.

Interpretation

The CAPM regression for the Energy industry portfolio produces a market beta of 0.8878. This suggests that the Energy portfolio moves positively with the market, but with slightly lower sensitivity than the overall market. A 1 percentage point increase in the market excess return is associated with an average 0.8878 percentage point increase in the Energy excess return. The estimated alpha is 0.1553% per month, but it is not statistically significant at the 5% level, as its p-value is 0.222. Therefore, the regression does not provide strong evidence of abnormal returns unexplained by market risk. The R-squared value is 0.539, meaning that approximately 53.9% of the variation in Energy excess returns is explained by the market excess return.

Statistic	Value	Meaning
Count	1198	
Mean	approximately 0	
Standard deviation	4.36	Typical monthly prediction error is about 4.36 percentage points
Minimum	-22.87	Worst negative miss
Maximum	22.73	Worst positive miss

Table 2. Descriptive Statistics of OLS Residuals: CAPM

Note: Residuals computed as actual minus predicted Energy excess return from the CAPM regression. Units are percentage points per month.

Learnings/Errors

Our output is **not correct**, but this is actually a very useful learning moment. The problem is here:

For monthly CAPM data, we should not have **4,792 observations**. We should have roughly **1,100–1,200 monthly observations**, depending on the latest data available. Your regression has accidentally included **extra tables** from the Kenneth French industry file, not just monthly industry returns.

Our earlier cleaning code kept all rows where the date looked like YYYYMM.

That means it accidentally included other monthly tables too, such as number of firms or firm size. That is why your *Enrgy* values became huge, and the intercept became:

const = 715.9295

That is not a return. That is a sign that non-return data entered the regression. After fixing it, you should expect something more sensible:

- No. Observations: around 1100–1200
- Mkt-RF coefficient: positive
- R-squared: clearly above 0
- const: small, probably close to 0

The Energy beta should usually be **positive**, because the Energy industry generally moves with the market.

Future Scope

For the next levels, we could add:

- SMB
- HML
- oil price changes
- inflation
- interest rates

2. CAPM vs Fama-French 3-Factor Model

Problem 2: Multiple Linear Regression — Comparing CAPM and Fama-French 3-Factor Models for the U.S. Energy Industry Portfolio

Research Anchor: Fama and French (1993), “Common Risk Factors in the Returns on Stocks & Bonds

We will expand the model by adding two more factors:

- SMB: Small minus Big; size factor
- HML: High minus Low; value factor

Does the Fama-French 3-Factor model explain Energy industry excess returns better than the simple CAPM model? Does adding more explanatory variables improve the model?

For this problem, we can reuse almost all the same dataset and cleaning code from Problem 1.

$Y = \text{Energy excess return}$

$X1 = \text{Market excess return}$

$X2 = \text{SMB}$

$X3 = \text{HML}$

$\text{Energy Excess Return} = \alpha + (\text{beta1} \times \text{Market Excess Return}) + (\text{beta2} \times \text{SMB}) + (\text{beta3} \times \text{HML})$

R-squared almost always increases when we add more variables. Adjusted R-squared is stricter.

Model	R-squared	Adjusted R-squared	AIC	BIC
CAPM	0.5393	0.5389	6928.56	6938.73
Fama-French 3-Factor	0.5748	0.5738	6836.47	6856.83

Table 3. Model Fit Comparison: CAPM vs. Fama-French 3-Factor Model

Note: Dependent variable is the monthly Energy industry excess return (Enrgy – RF). AIC and BIC computed using statsmodels OLS. Lower AIC/BIC indicates better model fit.

The Fama-French 3-Factor model is better because:

- R-squared increased
- Adjusted R-squared increased

- AIC decreased
- BIC decreased

Compared with CAPM, the market beta fell slightly. This is normal. Once we add SMB and HML, some of the variation previously attributed to the market is now explained by the other factors.

A negative -0.1586 SMB suggests that the Energy industry portfolio behaves more like large-cap stocks than small-cap stocks, or **it tends to move negatively with the small-firm factor**. Because the p-value is very small, this coefficient is statistically significant.

A positive HML coefficient of 0.3298 suggests that the **Energy industry portfolio has value-stock characteristics**. It tends to move positively with the value factor. This also makes economic sense because traditional energy companies often behave more like value/cyclical firms than high-growth technology firms.

The alpha has fallen. This means that once SMB and HML are added, **the unexplained average return becomes smaller**. But the alpha is still not statistically significant. There is no strong evidence that the Energy portfolio earned abnormal excess returns after controlling for market, size, and value factors.

Model	Residual Std. Dev.	Meaning
CAPM	4.3559	Typical monthly error around 4.36 percentage points
Fama-French 3-Factor	4.1847	Typical monthly error around 4.18 percentage points

Table 4. Residual Standard Deviation: CAPM vs. Fama-French 3-Factor Model

Note: Residual standard deviation measures the average magnitude of monthly prediction errors in percentage points. A lower value indicates a tighter model fit.

The residual mean is basically zero for both models, which is normal in OLS regression with an intercept.

FF3 residual standard deviation < CAPM residual standard deviation

So, the Fama-French 3-Factor model makes **smaller typical prediction errors** than CAPM.

Interpretation

The Fama-French 3-Factor model improves the explanation of Energy industry excess returns compared with the single-factor CAPM model. The R-squared increases from 0.5393 under CAPM to 0.5748 under the Fama-French 3-Factor model. The adjusted R-squared also increases from 0.5389 to 0.5738, suggesting that the additional explanatory power is meaningful even after accounting for the extra variables.

The market beta remains positive and statistically significant. In the Fama-French model, the market beta is 0.8682, meaning that a 1 percentage point increase in market excess return is associated with an approximately 0.8682 percentage point increase in Energy industry excess return, holding SMB and HML constant.

The SMB coefficient is negative and statistically significant, suggesting that the Energy industry portfolio behaves more like large-cap stocks than small-cap stocks. The HML coefficient is positive and statistically significant, suggesting that the Energy industry portfolio has value-stock characteristics.

The alpha decreases from 0.1553 in the CAPM model to 0.0787 in the Fama-French model. However, the alpha remains statistically insignificant. Therefore, there is no strong evidence of abnormal excess returns after controlling for market, size, and value factors.

Overall, the Fama-French 3-Factor model **provides a better fit than CAPM for explaining Energy industry excess returns.**

3. Temperature Shocks and Economic Growth

Problem 3: Hard Linear Regression —Temperature Shocks and Economic Growth

Research Anchor: Dell, Jones and Olken (2012), “Temperature Shocks and Economic Growth: Evidence from the Last Half Century”

This exercise uses a panel-style linear regression to examine the relationship between annual temperature and GDP per capita growth. The analysis is inspired by Dell, Jones and Olken (2012), using the `climate_panel.dta` file from the authors’ public replication dataset.

The paper studies whether hotter-than-usual years reduce economic growth, especially in poorer countries. The authors use historical fluctuations in temperature within countries to estimate the effect on economic outcomes, and they find that higher temperatures substantially reduce growth in poor countries.

The replication data is available through the AEA/openICPSR replication package. The package includes a file called *climate_panel.dta*, which is the main dataset we will use.

Dataset: Dell, Jones and Olken Replication Dataset — `climate_panel.dta`

We will learn three new ideas:

1. Panel data
2. Fixed effects
3. Interaction terms

Do hotter years reduce economic growth, and is this effect stronger in poor countries?

GDP_Growth = annual log growth of rgdpl

Poor_Country = 1 if initial income is below the sample median

Poor_Country = 0 otherwise

This is a beginner-friendly approximation of the paper’s poor-country split.

For our regression, we do not strictly need full country names. We only need a consistent country identifier for fixed effects.

Looking at the Scatter Plot, we observe that there is no clean downward-sloping or upward-sloping line that is obvious to the eye. That is why our simple model had a very low R-squared of 0.008. This means, Temperature alone explains less than 1% of the variation in GDP growth in the raw pooled data. That does **not** mean temperature is irrelevant. It means GDP growth is affected by many other things too.

Temperature is only one possible factor among many. So, the raw plot is naturally messy.

Why we needed fixed effects?

This raw plot compares very different countries with each other. For example:

- Cold rich countries
- Hot poor countries
- Cold poor countries
- Hot middle-income countries

But these countries differ in many permanent ways. That is why the raw scatter can be misleading. The fixed-effects model improves the logic by asking: *When the same country becomes hotter than usual in a particular year, does its GDP growth change?* This is the key idea of the Dell, Jones and Olken style analysis.

Why the poor-country interaction matters?

The raw plot combines richer and poorer countries together. But our regression found that the temperature effect is different across groups. So, the raw scatter hides the important result.

This plot tells us that the raw relationship between temperature and GDP growth is noisy and not visually obvious, but regression with fixed effects and poor-country interactions reveals a statistically significant negative temperature effect for poor countries.

The most important result is:

$$\text{Temperature} \times \text{Poor} = -1.2095$$

$$\text{p-value} = 0.000782$$

This means that the effect of temperature on GDP growth is significantly more negative for poor countries than for richer countries.

Variable	Coefficient	P-value	Interpretation
Temperature	0.2429	0.2989	Temperature effect for richer countries; not statistically significant
Precipitation	-0.0207	0.7113	Not statistically significant
Temperature × Poor	-1.2095	0.0008	Strongly significant negative additional effect for poor countries
Precipitation × Poor	0.0340	0.6693	Not statistically significant

Table 5. Panel Regression Results: Temperature, Precipitation, and GDP Growth (Problem 3)

Note: Dependent variable is annual GDP per capita growth. Country and year fixed effects included. *Poor_Country* = 1 if initial income below sample median. Standard errors clustered at country level. *** $p < 0.01$.

Group	Estimated temperature effect
Richer countries	0.2429
Poor countries	-0.9667

Table 6. *Estimated Temperature Effect on GDP Growth by Country Income Group (Problem 3)*

Note: Richer-country effect is the Temperature coefficient from the full specification. Poor-country total effect equals Temperature + Temperature $\times$ Poor. Units: percentage point change in annual GDP growth per 1°C increase in temperature.

For richer countries, a 1°C increase in annual temperature is associated with a 0.2429 percentage point increase in GDP growth, but this estimate is not statistically significant. So there is **no strong evidence of a temperature effect for richer countries in this specification.**

But for poor countries, a 1°C increase in annual temperature is associated with approximately a 0.9667 percentage point decrease in GDP growth. The estimated temperature effect is negative and economically meaningful. The p-Value figure for Poor countries confirms the **combined temperature effect for poor countries is statistically significant.**

Model Comparison:

Model	R-squared	Adjusted R-squared
Simple: Temperature only	0.0080	0.0079
Temperature + Precipitation	0.0097	0.0094
Country + Year Fixed Effects	0.0898	0.0577
Fixed Effects + Poor Interactions	0.0916	0.0593

Table 7. *Model Fit Progression: R-Squared Across Specifications (Problem 3)*

Note: Each row represents a sequentially more complex regression specification. Country and year fixed effects are included in the last two rows. R-squared for fixed-effects models is within-variation R-squared.

The simple model explains very little variation in GDP growth: as it has R-squared of 0.0080. That is not surprising. GDP growth is affected by many things: investment, trade, politics, conflict, global shocks, financial crises, institutions, productivity, etc.

But once we add country and year fixed effects, the model improves: R-squared increases from 0.0097 to 0.0898. Then adding the poor-country interaction improves it slightly further.

The improvement in R-squared is not huge, but the **Temperature × Poor coefficient is statistically significant**, which is the key result.

The coefficient is -0.9667 . This means that for poor countries, a 1°C increase in annual temperature is associated with approximately a **0.97 percentage point decrease in GDP per capita growth**.

Combined Temperature Effect for Poor Countries: To test whether the total temperature effect for poor countries is statistically significant, I tested the joint expression: $\text{Temperature} + \text{Temperature} \times \text{Poor} = 0$. The estimated combined effect is -0.9667 , with a p-value of 0.002. This means that for poor countries, a 1°C increase in annual temperature is associated with approximately a 0.97 percentage point reduction in GDP per capita growth.

The 95% confidence interval ranges from -1.586 to -0.348 . Since the entire interval is below zero, the result provides evidence that temperature shocks have a statistically significant negative effect on economic growth in poor countries. This finding is consistent with the broad conclusion of Dell, Jones and Olken that poor countries are more vulnerable to higher temperatures.

Results are meaningful and they match the broad story of the Dell, Jones and Olken paper: **temperature shocks appear more damaging for poorer countries**.

Interpretation

This exercise estimates the relationship between annual temperature and GDP per capita growth using country-year panel data. The analysis is inspired by Dell, Jones and Olken's study of temperature shocks and economic growth.

The simple temperature-only model has very low explanatory power, with an R-squared of 0.008. This suggests that temperature alone explains only a small portion of the variation in annual GDP growth across countries and years. Adding precipitation increases the R-squared slightly to 0.0097.

The model improves after including country and year fixed effects. The R-squared rises to 0.0898, indicating that controlling for permanent country-level differences and common global year shocks helps explain more of the variation in GDP growth.

The main specification adds interaction terms between climate variables and the poor-country dummy. In this model, the coefficient on Temperature is 0.2429, but it is not statistically significant. This means there is no strong evidence that annual temperature has a clear effect on GDP growth for richer countries in this specification.

The coefficient on Temperature $\times$ Poor is -1.2095 and statistically significant, with a p-value of 0.0008. This suggests that the temperature-growth relationship is significantly more negative for poor countries. Combining the Temperature coefficient with the Temperature $\times$ Poor coefficient gives an estimated temperature effect of approximately -0.9667 for poor countries. Therefore, a 1°C increase in annual temperature is associated with about a 0.97 percentage point reduction in GDP per capita growth for poor countries.

Overall, the results support the broad conclusion that poorer countries are more economically vulnerable to temperature shocks than richer countries.

Conclusion

This exercise finds that the relationship between temperature and GDP growth differs across richer and poorer countries. The temperature coefficient for richer countries is positive but statistically insignificant, meaning there is no strong evidence of a temperature-growth effect for richer countries in this specification.

However, the interaction between temperature and the poor-country dummy is negative and statistically significant. The combined estimated temperature effect for poor countries is -0.9667, suggesting that a 1°C increase in annual temperature is associated with nearly a 1 percentage point reduction in GDP per capita growth. Therefore, the results support the idea that poorer countries are more economically vulnerable to temperature shocks.

References and Bibliography

Fama, E.F. and French, K.R. (1993) 'Common risk factors in the returns on stocks and bonds', *Journal of Financial Economics*, 33(1), pp. 3–56.

French, K.R. (2026) *Kenneth R. French Data Library*. Dartmouth College. Available at: Kenneth R. French Data Library.

Dell, M., Jones, B.F. and Olken, B.A. (2012) 'Temperature shocks and economic growth: Evidence from the last half century', *American Economic Journal: Macroeconomics*, 4(3), pp. 66–95. DOI: 10.1257/mac.4.3.66.

Dell, M., Jones, B.F. and Olken, B.A. (2012) *Replication data for: Temperature Shocks and Economic Growth: Evidence from the Last Half Century: AEJ Mac2010 0092 reproduction: climate_panel.dta*. Nashville, TN: American Economic Association [publisher]. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor], 2019-10-12. DOI: 10.3886/E114251V1-99620.

Appendix

A.1 Problem 1: CAPM Beta Estimation (Energy Industry Portfolio)

OLS Regression Results						
=====						
Dep. Variable:	Energy_Excess	R-squared:	0.539			
Model:	OLS	Adj. R-squared:	0.539			
Method:	Least Squares	F-statistic:	1400.			
Date:	Sat, 20 Jun 2026	Prob (F-statistic):	1.63e-203			
Time:	23:39:02	Log-Likelihood:	-3462.3			
No. Observations:	1198	AIC:	6929.			
Df Residuals:	1196	BIC:	6939.			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.1553	0.127	1.223	0.222	-0.094	0.404
Mkt-RF	0.8878	0.024	37.418	0.000	0.841	0.934
=====						
Omnibus:	122.282	Durbin-Watson:	1.847			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	454.567			
Skew:	0.441	Prob(JB):	1.96e-99			
Kurtosis:	5.886	Cond. No.	5.40			
=====						

Figure 1. CAPM scatter plot of Energy industry excess returns against market excess returns. The fitted line represents the OLS regression; each point is a monthly observation.

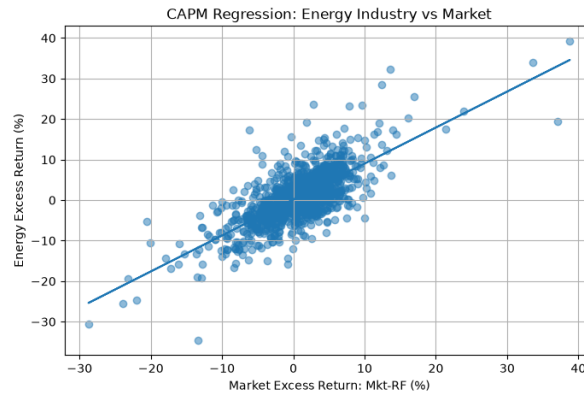


Figure 2. OLS regression output for the CAPM model (Energy excess return on market excess return). Key statistics: $\beta = 0.8878$, $\alpha = 0.1553$ ($p = 0.222$), $R^2 = 0.539$, $n = 1,198$.

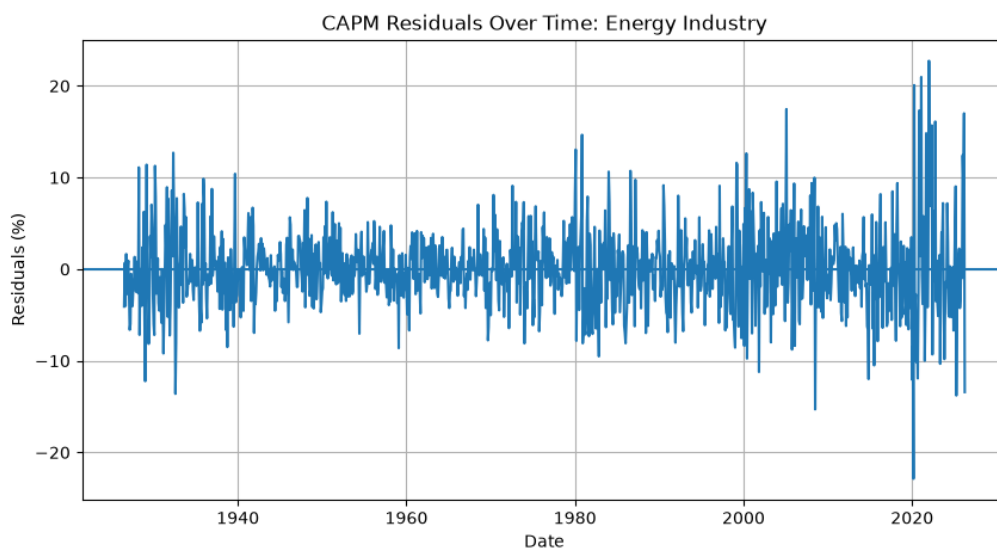


Figure 3. Residual plot for the CAPM model. Residuals are plotted against time (monthly). Spikes indicate periods where market risk alone fails to explain Energy returns.

A.2 Problem 2: CAPM vs. Fama-French 3-Factor Model

OLS Regression Results						
=====						
Dep. Variable:	Enrgy_Excess	R-squared:	0.575			
Model:	OLS	Adj. R-squared:	0.574			
Method:	Least Squares	F-statistic:	538.1			
Date:	Sun, 21 Jun 2026	Prob (F-statistic):	3.78e-221			
Time:	00:36:14	Log-Likelihood:	-3414.2			
No. Observations:	1198	AIC:	6836.			
Df Residuals:	1194	BIC:	6857.			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0787	0.122	0.643	0.520	-0.161	0.319
Mkt-RF	0.8682	0.025	35.365	0.000	0.820	0.916
SMB	-0.1586	0.041	-3.899	0.000	-0.238	-0.079
HML	0.3298	0.035	9.408	0.000	0.261	0.399
=====						
Omnibus:	112.245	Durbin-Watson:	1.857			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	481.004			
Skew:	0.338	Prob(JB):	3.56e-105			
Kurtosis:	6.030	Cond. No.	5.66			
=====						

Figure 4. OLS regression output for the Fama-French 3-Factor model (Energy excess return on Mkt-RF, SMB, HML). Market beta = 0.8682; SMB coefficient = -0.1586 ; HML coefficient = 0.3298 ; $R^2 = 0.5748$.

A.3 Problem 3: Temperature Shocks and Economic Growth

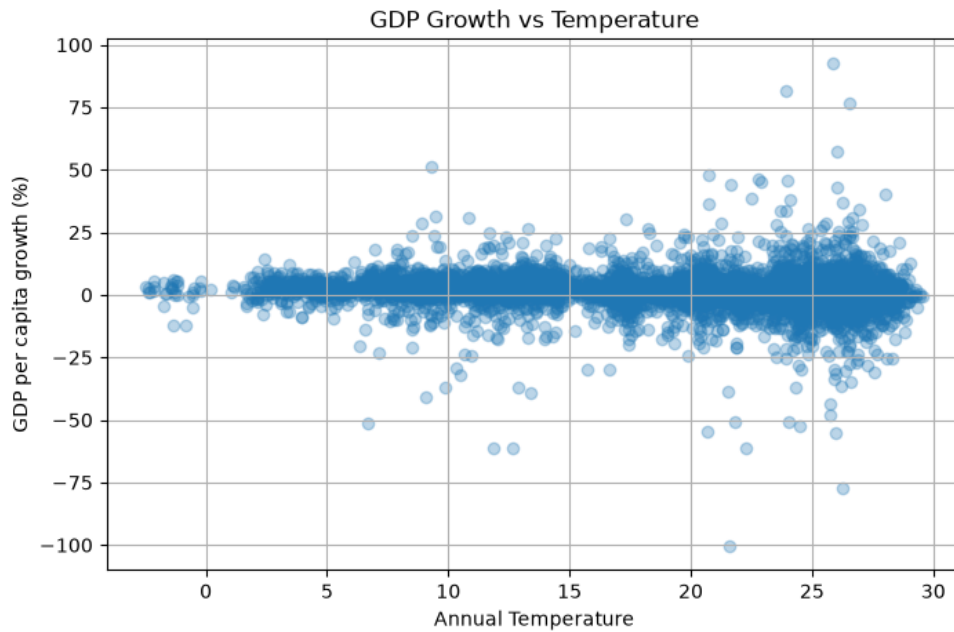


Figure 5. Raw scatter plot of annual temperature against GDP per capita growth across all country-years. The near-zero slope reflects substantial cross-sectional heterogeneity masked before fixed effects are applied.

OLS Regression Results						
=====						
Dep. Variable:	GDP_Growth		R-squared:		0.008	
Model:	OLS		Adj. R-squared:		0.008	
Method:	Least Squares		F-statistic:		51.87	
Date:	Sun, 21 Jun 2026		Prob (F-statistic):		6.62e-13	
Time:	02:28:37		Log-Likelihood:		-21898.	
No. Observations:	6426		AIC:		4.380e+04	
Df Residuals:	6424		BIC:		4.381e+04	
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	3.4252	0.257	13.328	0.000	2.921	3.929
Temperature	-0.0909	0.013	-7.202	0.000	-0.116	-0.066
=====						
Omnibus:	2085.884		Durbin-Watson:		1.708	
Prob(Omnibus):	0.000		Jarque-Bera (JB):		183544.863	
Skew:	-0.604		Prob(JB):		0.00	
Kurtosis:	29.154		Cond. No.		57.5	
=====						

Figure 6. Panel regression output including country and year fixed effects but without poor-country interaction terms. Estimates the average temperature-growth relationship after removing permanent country differences.

OLS Regression Results						
=====						
Dep. Variable:	GDP_Growth	R-squared:	0.010			
Model:	OLS	Adj. R-squared:	0.009			
Method:	Least Squares	F-statistic:	31.38			
Date:	Sun, 21 Jun 2026	Prob (F-statistic):	2.76e-14			
Time:	02:29:17	Log-Likelihood:	-21893.			
No. Observations:	6426	AIC:	4.379e+04			
Df Residuals:	6423	BIC:	4.381e+04			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	3.2274	0.264	12.236	0.000	2.710	3.744
Temperature	-0.1074	0.014	-7.912	0.000	-0.134	-0.081
Precipitation	0.0438	0.013	3.287	0.001	0.018	0.070
=====						
Omnibus:	2070.185	Durbin-Watson:	1.711			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	184064.319			
Skew:	-0.587	Prob(JB):	0.00			
Kurtosis:	29.193	Cond. No.	69.2			
=====						

Figure 7. Full panel regression with poor-country interaction terms. The Temperature $\times$ Poor coefficient (-1.2095 , $p < 0.001$) identifies the differential temperature effect for low-income countries.

Test for Constraints						
=====						
	coef	std err	t	P> t	[0.025	0.975]

c0	-0.9667	0.316	-3.062	0.002	-1.586	-0.348
=====						

Figure 8. 95% confidence interval for the combined temperature effect in poor countries (Temperature + Temperature $\times$ Poor = -0.9667). The entire interval lies below zero, confirming statistical significance.